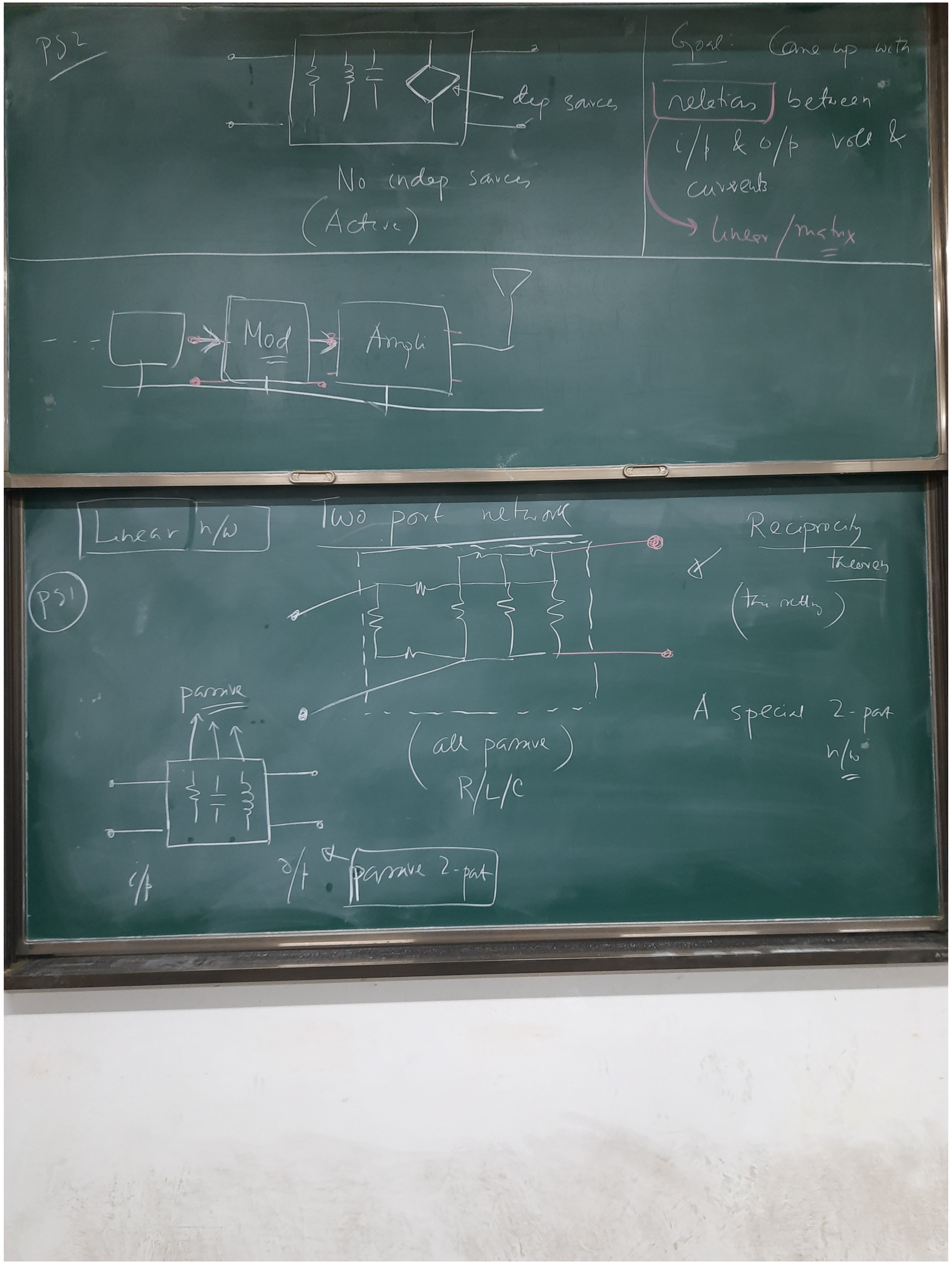

PS2
dep sources
Mod
Ampli
PS1
passive
i/p
o/p

No indep sources
(Active)

Goal: Come up with relations between i/p & o/p volt & currents
→ linear / matrix

Linear n/w

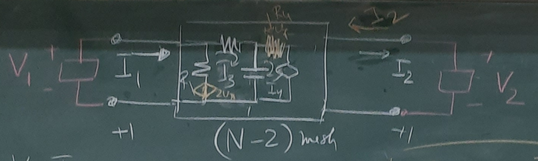
Two port network

(all passive)
R/L/C

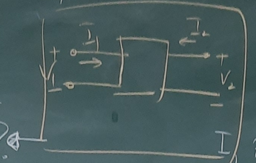
✗ passive 2-port

Reciprocity theorem
(this netting)

A special 2-port n/w


V_1
I_1
R_1
I_3
I_4
$2v_x$
R_4
I_2
V_2
+1
(N − 2) mesh
+1

$V_1 \; I_1$	$V_2 \; I_2$
$V_1 \; V_2$	$I_1 \; I_2$?
$I_1 \; I_2$	$V_1 \; V_2$?
$V_1 \; I_1$	$V_2 \; I_2$?



Coeff =

$-R_1$

$Z_{14} = -2R_4$

$$I_1 (Z_{11}) + I_3 Z_{13} + I_N Z_{1N} + I_2 (Z_{12}) = V_1$$

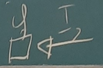
$Z_{15} = 0$

$I_1 R_1 - I_3 R_1 = V_1$

$+ I_4 R_4 \cdot 2$

i/p

1st mesh $\sum_j Z_{1j} I_j = V_1$ | o/p mesh $\sum_j Z_{2j} I_j = V_2$



PS3

PS4

$\sum_j Z_{ij} I_j = 0$ ← No indep source

inside → ☐

$$\underbrace{\begin{bmatrix} Z_{11} & Z_{12} & Z_{13} & \cdot & Z_{1N} \\ Z_{21} & Z_{22} & & & \\ \vdots & & & & \\ \vdots & & & & \\ Z_{N1} & \cdots & & & Z_{NN} \end{bmatrix}}_{[Z]} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

25

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix} = [Z]^{-1} \begin{bmatrix} V_1 \\ V_2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Suppose it was a 1-port n/w

(No RED leads)

$\frac{V_1}{I_1}$ ~ input resistance Z_{in}

Recall $Z(s) = \frac{V(s)}{I(s)}$ (Cap / Ind s-domain)

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} 2nd \\ 3rd \\ \vdots & Z_{(-2)} \\ Nth \end{bmatrix}^{-1} \begin{bmatrix} V_1 \\ \cancel{V_2} \\ \vdots \\ 0 \end{bmatrix}$$

2nd row

$(N-1 \times N-1)$ ∅

$$= \begin{bmatrix} Z_{(-2)}^{-1} \end{bmatrix} \begin{bmatrix} V_1 \\ 0 \\ \vdots \end{bmatrix}$$

$$I_1 = \left[Z_{(-2)}^{-1}\right]_{11} V_1$$

26

$$Z = \begin{bmatrix} \boxed{z_{11}} & z_{12} & \cdots \\ z_{21} & & \\ \vdots & \Delta_{11} & \\ \vdots & & \end{bmatrix}$$

$$\left[Z^{-1}\right]_{11} = \frac{|\Delta_{11}|}{|Z|}$$

Cramer's law:

$$I_1 = \left[Z^{-1}\right]_{11} V_1 + \left[Z^{-1}\right]_{12} V_2$$

$$I_2 = \left[Z^{-1}\right]_{21} V_1 + \left[Z^{-1}\right]_{11} V_2$$

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ \vdots \end{bmatrix} = \left[Z^{-1}\right] \begin{bmatrix} V_1 \\ V_2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \underbrace{\left[Z^{-1}\right]}_{2 \times 2} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

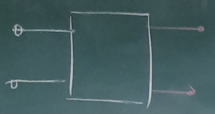
$$\begin{bmatrix} Z^{-1}_{11} & Z^{-1}_{12} & Z^{-1}_{1N} \\ Z^{-1}_{21} & Z^{-1}_{22} & \vdots \\ \vdots & & \end{bmatrix}$$

1st 2 rows & columns

PS 7

$$I_1 = y_{11} V_1 + y_{12} V_2$$

$$I_2 = y_{21} V_1 + y_{22} V_2$$



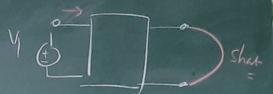
Suppose $V_2 = 0$

$$\Rightarrow y_{11} = \boxed{\frac{I_1}{V_1}}\Bigg|_{V_2=0}$$

i/p admittance

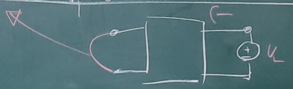
Short circuit

(o/p port)
Short-circuit
input admittance



$$y_{22} = \frac{I_2}{V_2}\Bigg|_{V_1=0}$$

Short circuit o/p admittance



y_{12} / y_{21} are transfer admittance

$$y_{12} = \frac{I_1}{V_2}\Bigg|_{V_1=0} \qquad y_{21} = \frac{I_2}{V_1}\Bigg|_{V_2=0}$$